

The core activities in this unit total about 25.5 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Part 1 Why do we use the materials we do?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	It's a material world	AC SIS140	Engage	Students are asked to identify the properties of the materials used in the Rock, Paper, Scissors game and explore the properties and used of materials around them.	Core
			Hands-on, Classroom & Digital		
			Medium		
1.2	What is the property?	AC SSU151	Explore & Explain	Students examine groups of materials to determine their common properties and link them to the correct scientific words.	Core
			Hands-on		
			Medium		
1.3	Metals and their properties	AC SSU151 AC SHE134 AC SHE136 AC SIS141 AC SIS144 AC SIS145 AC SIS234	Explore	Students will examine a range of metals to explore their key properties. They will go on to consider ways in which metals are used in modern societies.	Core
			Hands-on & Digital		
			Medium		
1.4	Rock to scissors	AC SSU151	Elaborate & Evaluate	Students will use the Student Digital to discover what minerals are mined near them and research the properties and uses of iron.	Optional
			Classroom & Digital		
			Medium		

Part 2 What is the matter?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	Investigating the properties of solids, liquids and gases	ACSSU151 ACSIS139	Explore	Students will conduct a series of experiments to discover some of the properties of solids, liquids and gases.	Core
			Hands-on		
			Medium		
2.2	Tiny particles	ACSSU151 ACSHE134 ACSHE226 ACSIS144	Explore & Explain	Students will use a learning object to explore the behaviour of particles in each state of matter and then demonstrate their understanding by using the particle motion to explain the Brownian motion of particles in milk observed under the microscope.	Core
			Hands-on & Digital		
			Medium		
2.3	Slime – what is its state?	ACSSU151	Elaborate	Students make slime (corn flour) and investigate its properties. They will try to classify slime based on what they know about the three states of matter. They will explore this special type of fluid.	Optional
			Hands-on		
			Medium		
2.4	What state is that?	ACSSU151	Explore & Elaborate	Students apply what they have learnt about the three states of matter to everyday materials.	Optional
			Classroom & Digital		
			Medium		
2.5	Gas detective	ACSSU151	Explore & Elaborate	Students conduct a series of inquiries about gases, using household items and then choose one of three topics for further investigation.	Optional
			Hands-on		
			Medium		
2.6	Two more states that matter	ACSSU151	Elaborate	Students discover two more states of matter: plasma and Bose-Einstein condensate.	Optional
			Digital		
			Short		

Part 3 Elements, compounds and mixtures

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	Is it pure or a mixture?	ACSSU152 ACSIS140 ACSIS141	Engage, Explore, Explain & Elaborate	Students explore the differences between a pure substance and a mixture through some sandcastle play. Students revise their understanding of mixtures, pure substances, elements and compounds and sort substances accordingly.	Core
			Hands-on & Digital		
			Long		
3.2	Separating mixtures	ACSSU152 ACSIS140	Elaborate	Students will use their knowledge of the properties of substances to separate a mystery mixture.	Core
			Hands-on & Digital		
			Short		
3.3	What are alloys?	ACSSU152	Elaborate	Students investigate different useful alloys and determine how their elements affect their properties.	Core
			Digital		
			Short		
3.4	Do the powders contain the same particles?	ACSSU152 ACSIS139	Explore & Evaluate	Students investigate the properties of three copper compounds to discover the differences between elements and compounds.	Core
			Hands-on		
			Long		
3.5	Splitting water	ACSSU152 ACSIS140	Explore	Students electrolyse water using basic equipment.	Core
			Hands-on		
			Medium		
3.6	Elements: the simplest substances	ACSSU152	Explain	Students will explore the range of elements and their properties using an interactive periodic table.	Core
			Digital		
			Short		

Part 3 Elements, compounds and mixtures (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.7	The purity of water	ACSSU152	Explore & Explain	Students use models to learn about the difference between elements, compounds and mixtures.	Core
			Classroom		
			Medium		
3.8	The periodic table	ACSSU152	Elaborate	Students carry out an activity on-line to simulate Mendeleev's analysis of six elements to produce the very first fragment of the periodic table.	Core
			Digital		
			Short		
3.9	Iron – element, compound and mixture	ACSSU152	Elaborate	Students will investigate some uses of the element iron, and compounds and mixtures of iron.	Core
			Classroom		
			Short		
3.10	Is it an element, compound or mixture?	ACSSU152	Evaluate	Students will evaluate their understanding of PART 3 . They will identify elements, compounds and mixtures, and the type and number of elements in a compound.	Core
			Classroom		
			Short		

Part 4 Changing state: a physical change

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Observing changes in water	ACSSU151 ACSIS141 ACSIS144 ACSIS148	Explore & Explain	Students investigate the changes that take place when water moves from a solid □ liquid □ gas, and identify the correct terms of those changes. They also identify these changes as examples of a 'physical change'.	Core
			Hands-on & Digital		
			Long		
4.2	Ice play	ACSSU151 ACSHE135 ACSHE136	Explore	Students follow a recipe for making ice cream. The recipe utilises the effect of salt on the melting point of water to lower the freezing point.	Core
			Hands-on		
			Medium		
4.3	Can matter skip states?	ACSSU151	Engage & Explore	Students see examples of substances that change from a solid, directly to a gas (sublimation). They also learn about uses of sublimation.	Core
			Digital		
			Short		
4.4	Detecting an invisible gas	ACSSU151 ACSSU152	Explore & Explain	Students use a simple test to detect the presence of carbon dioxide gas. They go on to complete a digital activity familiarising them with a number of gases that they may come across in everyday life.	Optional
			Hands-on & Digital		
			Short		
4.5	What is happening to me?	ACSSU151	Elaborate & Evaluate	Students imagine they are water molecules in an ice block and describe what happens to them when left in the sun.	Core
			Classroom		
			Short		

Part 5 Chemical change

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	Some chemical changes	ACSSU225 ACSIS145	Engage	Students conduct a series of experiments, observing different examples of chemical reactions.	Core
			Hands-on		
			Medium		
5.2	Observing reactions	ACSSU225 ACSIS145	Explore & Explain	Students observe a further range of chemical reactions to extend their knowledge of observations linked to chemical changes.	Core
			Hands-on		
			Medium		
5.3	Reactants and products	ACSSU225 ACSIS140	Explore & Explain	Students conduct a series of experiments to observe chemical reactions. They will identify the reactants and products, and write the word equations for the reactions.	Core
			Hands-on		
			Medium		
5.4	Some natural reactions	ACSSU225	Explain	Students discover some natural reactions – photosynthesis and corrosion.	Core
			Digital		
			Medium		
5.5	Investigating some useful reactions	ACSSU225 ACSHE134 ACSHE136 ACSIS146	Elaborate	Students can make one or more of the following polymers – plastic, glue or rubber.	Core
			Hands-on		
			Long		
5.6	A rusty problem	ACSSU225 ACSIS141	Explore & Explain	Students conduct investigations to answer the question 'What is rust?'	Core
			Hands-on		
			Long		
5.7	Preventing rust	ACSSU225 ACSHE135 ACSIS141	Elaborate & Evaluate	Students conduct an investigation to answer 'How do we prevent the corrosion of iron?' They will also investigate how sacrificial anodes prevent rusting.	Core
			Hands-on		
			Medium		

Part 5 Chemical change (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.8	Do other metals corrode like iron?	ACSSU225 ACSIS140 ACSIS141 ACSIS145	Elaborate	Students design and conduct investigations to answer the question ' <i>Do other metals corrode like iron?</i> '	Core
			Hands-on		
			Long		
5.9	Chemical hazards	ACSHE134 ACSHE135 ACSIS140	Elaborate	Students watch short video clips to reflect on the impact of toxic particles in the air and consider safety precautions when dealing with chemicals.	Core
			Digital		
			Short		
5.10	Sample test		Evaluate	A sample summative test (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Classroom		
			Medium		

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